07 – Binary systems, electronics and microprocessors

If it is not true then, it is false…

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OUTLINE

1) Two-state systems
2) Boolean Algebra
3) Logic circuits
4) Microprocessors

1) TWO-STATE SYSTEMS

- The binary system works with two states: 0/1, false/true, off/on, open/closed, low/high
- An electric cable can have or not an electric current going through it
- When the Amplitude is maximum: 1
- When the Amplitude is minimum: 0

1) TWO-STATE SYSTEMS

- With one cable, you can represent ___ states: 0 or 1
- With two cables, you can represent ___ states:
- With three cables, you can represent ___ states:
- …
- With N cables, you can represent ___ states:

1) TWO-STATE SYSTEMS

MSB **LSB**

$$\underbrace{1\ 0\ 1\ 1\ 0\ 1\ 0\ 1}_{\text{BINARY}} = \underbrace{\qquad\qquad}_{\text{DECIMAL}}$$

1) TWO-STATE SYSTEMS

- Introducing a new notation with the base as subscript:

$$(1\ 1\ 0\ 0\ 1)_2$$ BINARY

$$(1\ 1\ 0\ 0\ 1)_{10}$$ DECIMAL

1) TWO-STATE SYSTEMS

- Size of binary numbers:
  - 8 bits, 256 possible values (2^8)
  - 16 bits, 65 536 possible values (2^{16})
  - 32 bits, 4 294 967 396 possible values (2^{32})
  - 64 bits, 18 446 744 073 709 551 616 possible values (2^{64})

1) TWO-STATE SYSTEMS

- Names of binary numbers:
  - 1 binary digit: 1 bit (**bi**nary digi**t**)
  - 4 bits : a quad
  - 8 bits : a byte
  - 16 bits : a word
  - 32 bits : a double word
  - 64 bits : a quad word

1) TWO-STATE SYSTEMS

- How do you know if a binary number is odd or even? (at first glance)

Odd or Even?

- $(1\ 0\ 1\ 1)_2 =$
- $(10\ 1\ 0\ 1\ 1)_2 =$
- $(1\ 0\ 1\ 1\ 1\ 0)_2 =$
- $(1\ 0\ 1\ 0\ 1\ 1\ 0\ 1)_2 =$
- $(1\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 1\ 0\ 1\ 1)_2 =$
- $(1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1\ 0\ 0)_2 =$

1) TWO-STATE SYSTEMS

- What happen if you shift a binary number one position to the left?
- $(1\ 0\ 1\ 1)_2 =$
- $(1\ 0\ 1\ 1\ 0)_2 =$
- What happen if you shift a binary number one position to the right?
- $(1\ 0\ 1\ 0)_2 =$
- $(1\ 0\ 1)_2 =$
- What happen if you shift a binary number N positions to the left?
- What happen if you shift a binary number N positions to the right?

2) BOOLEAN ALGEBRA

- In 1854, George Boole defines boolean algebra to compute and to use functions with binary numbers
- It is based on three basic functions:
  - AND
  - OR
  - NOT

2) BOOLEAN ALGEBRA

- Truth tables:

Source: https://en.wikipedia.org/wiki/Boolean_algebra

$\overline{a}$

a	NOT a
0	
1	

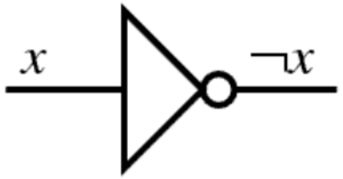
a.b

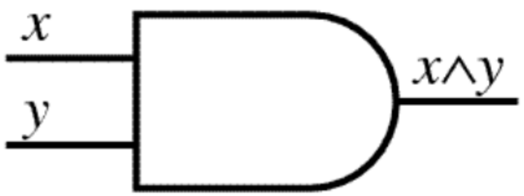
a	b	a AND b
0	0	
0	1	
1	0	
1	1	

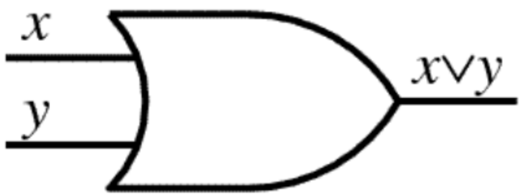
a+b

a	b	a OR b
0	0	
0	1	
1	0	
1	1	

Logic gates:







2) BOOLEAN ALGEBRA

- Laws:

	AND	OR
Identity	$1.a =$	$0+a =$
Annihilator	$0.a =$	$1+a =$
Idempotence	$a.a =$	$a+a =$
Complementation	$a.\overline{a} =$	$a+\overline{a} =$
Commutativity	$a.b =$	$a+b =$
Associativity	$a.(b.c) =$	$a+(b+c) =$
Absorption	$a.(a+b) =$	$a+(a.b) =$
Distributivity	$a.(b+c) =$	$a+(b.c) =$

2) BOOLEAN ALGEBRA

- Other laws:

De Morgan laws:

$\overline{\overline{a}} =$ $\overline{a.b} =$ $\overline{a+b} =$

Exclusive OR:

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

$$a \oplus b = (a + b).\overline{(a.b)}$$
$$a \oplus b = (a.\bar{b}) + (\bar{a}.b)$$
$$a \oplus b = \overline{((a.b) + (\bar{a}.\bar{b}))}$$
$$a \oplus b = (a + b).(\bar{a} + \bar{b})$$

2) BOOLEAN ALGEBRA

- Not OR (NOR):

a	b	$\overline{(a+b)}$
0	0	1
0	1	0
1	0	0
1	1	0

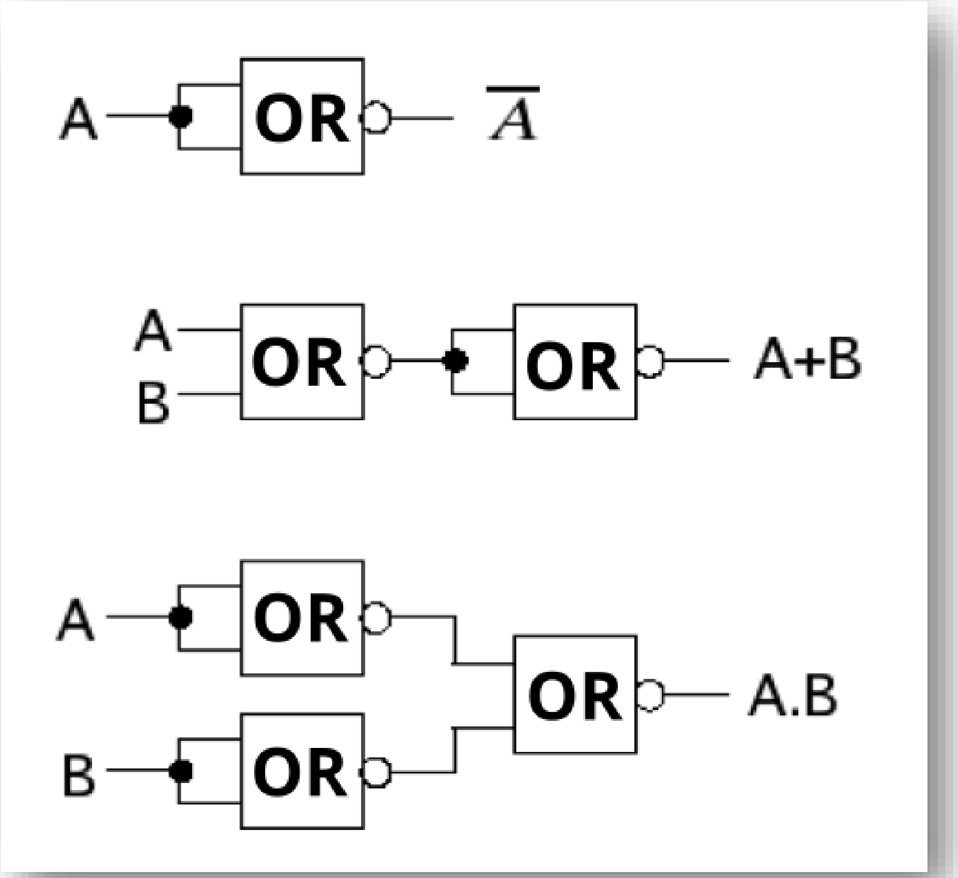
Not AND (NAND):

a	b	$\overline{(a.b)}$
0	0	1
0	1	1
1	0	1
1	1	0

Theorem: **EVERY** logic function can be designed from **NOR** and **NAND** logical operators!

2) BOOLEAN ALGEBRA

- Examples of boolean functions made with NOR gates only:



3) LOGIC CIRCUITS

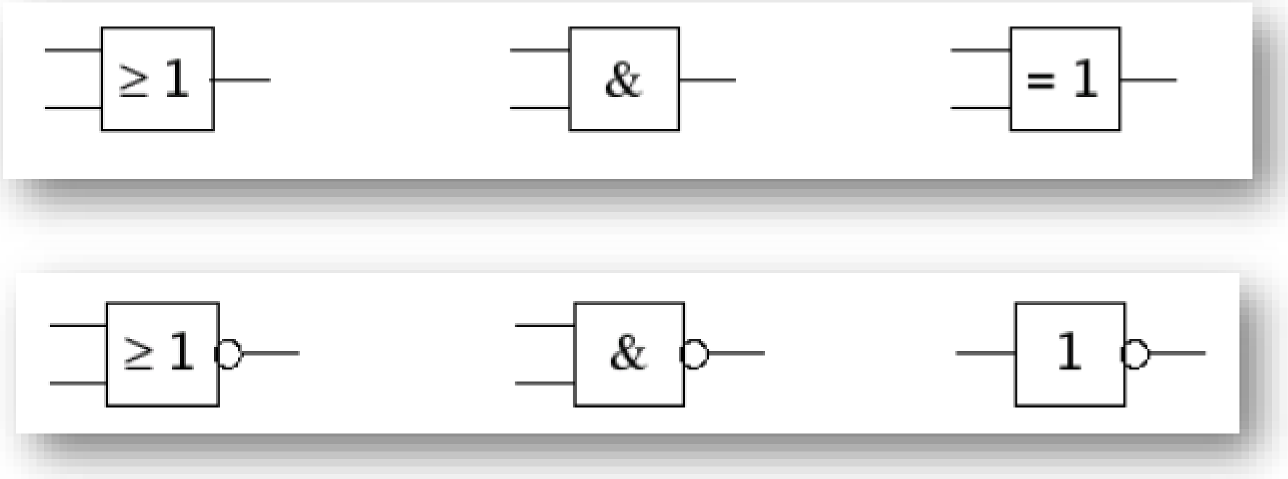
- Inside the microprocessor, in the control unit and in the Arithmetic and Logic Unit (ALU), electronic circuits are used to create registers, addition, subtraction, multiplication, division, logical operations…
- A logic circuit is an electronic circuit that is dedicated to create a logic function

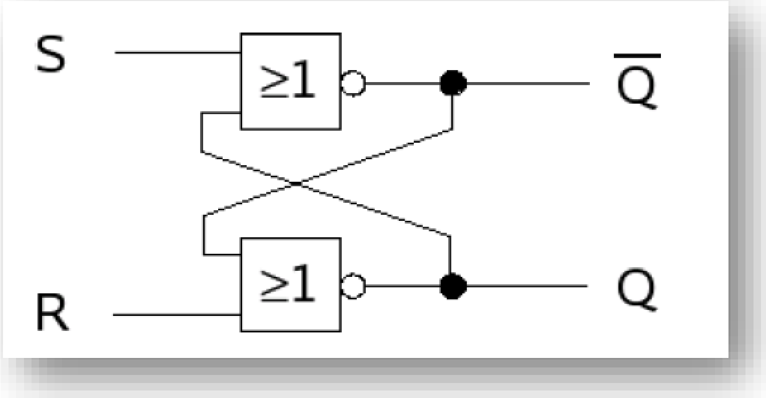
$(y_1, \ldots, y_m) = f(x_1, \ldots, x_n)$

There is always a delay between the change of the inputs and the update of the outputs

3) LOGIC CIRCUITS

- There are two types of logic circuits:
  - **Combinational logic** circuits in which the outputs depend only on the inputs (no internal memorization)
  - **Sequential logic** circuits in which the outputs depend on the inputs and on the previous state of the outputs (internal memorization)



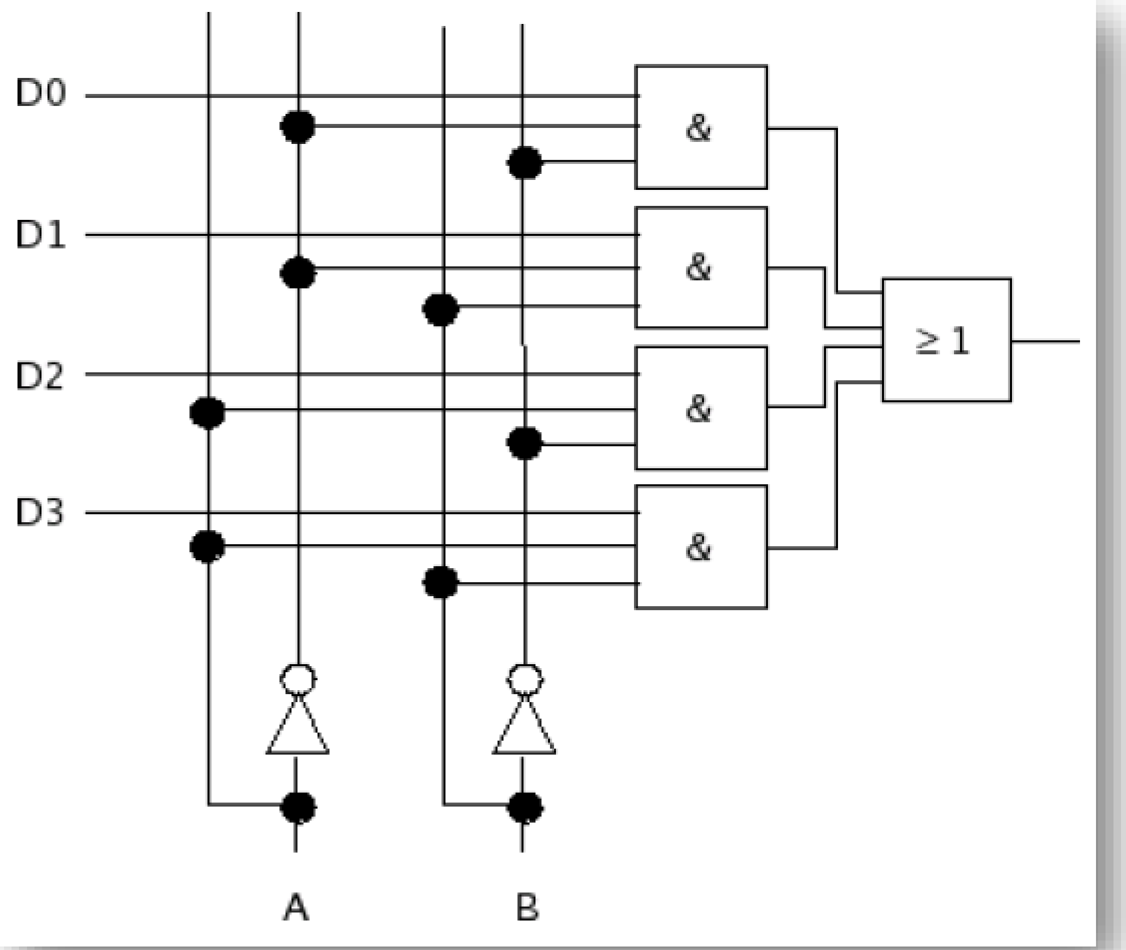


Combinatorial **Sequential**

Source: https://en.wikipedia.org/wiki/Combinational_logic, https://en.wikipedia.org/wiki/Sequential_logic

3) LOGIC CIRCUITS

- Multiplexer:
  - 2^N inputs
  - 1 output
  - N selection inputs

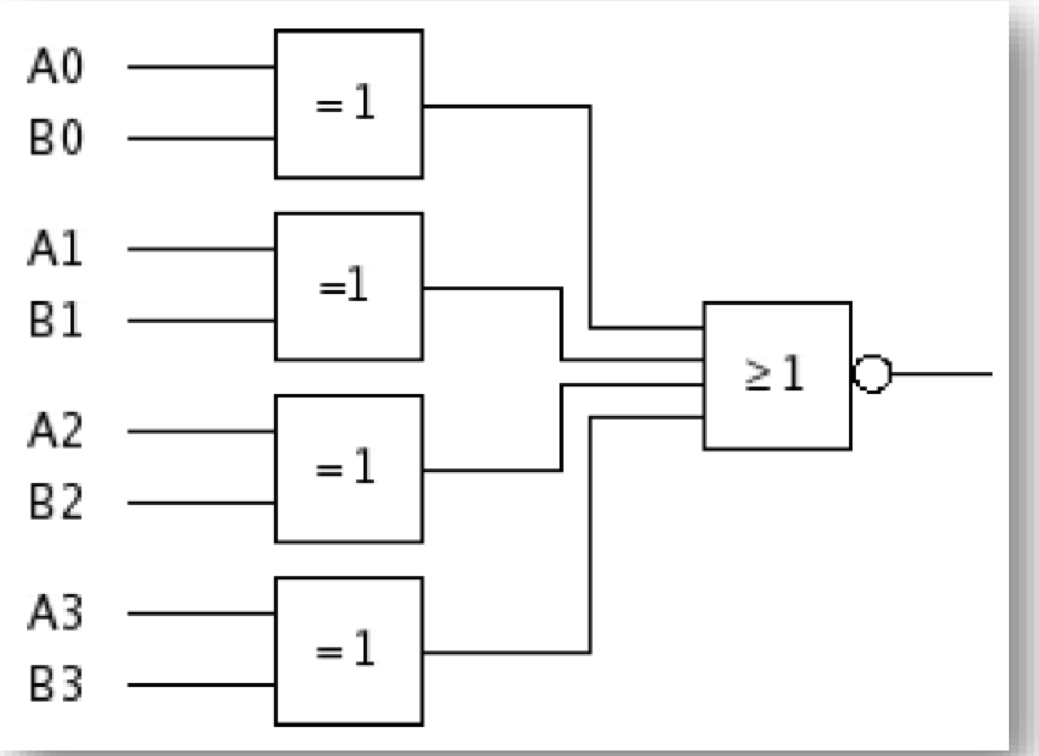


Combinatorial

Source: https://en.wikipedia.org/wiki/Multiplexer

3) LOGIC CIRCUITS

- Digital Comparator:
  - 2N inputs
  - 1 output
  - Used to compare two binary numbers a, b
  - Output = 1 if a == b
  - Output = 0 else

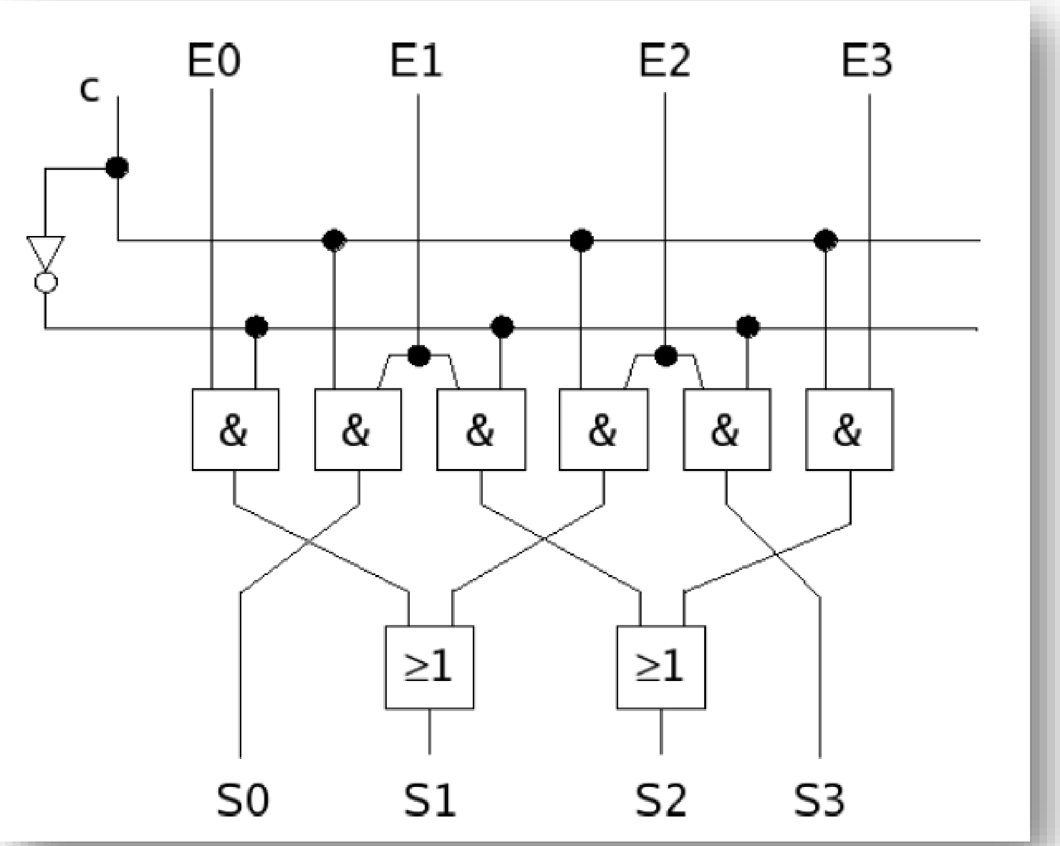


Combinatorial

Source: https://en.wikipedia.org/wiki/Multiplexer

3) LOGIC CIRCUITS

- Barrel shifter:
  - 1 command input c
  - 1 input number E (here 4 bits)
  - 1 output number S (here 4 bits)
  - It shifts a binary number to the left or the right depending on the value of c
  - (It is a multiplication or division by 2)



Combinatorial

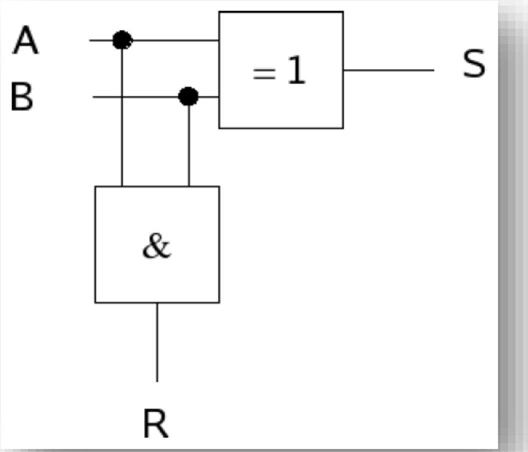
Source: https://en.wikipedia.org/wiki/Barrel_shifter

3) LOGIC CIRCUITS

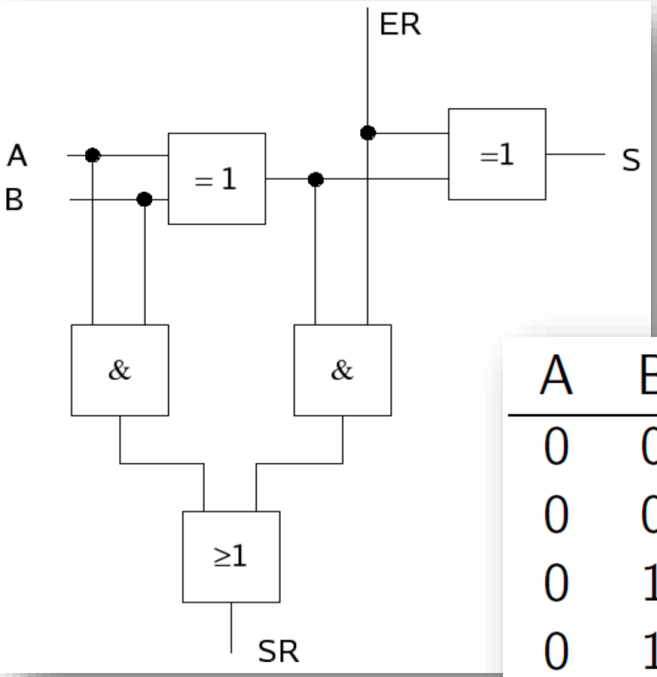
- Adder (half-adder and full-adder):

- Adds two numbers with carry information

- Adds two numbers with full result



A	B	S	R
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



A	B	ER	S	SR
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Combinatorial

Source: https://en.wikipedia.org/wiki/Barrel_shifter

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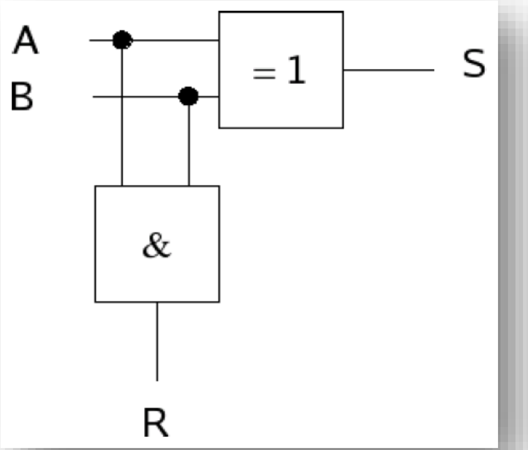
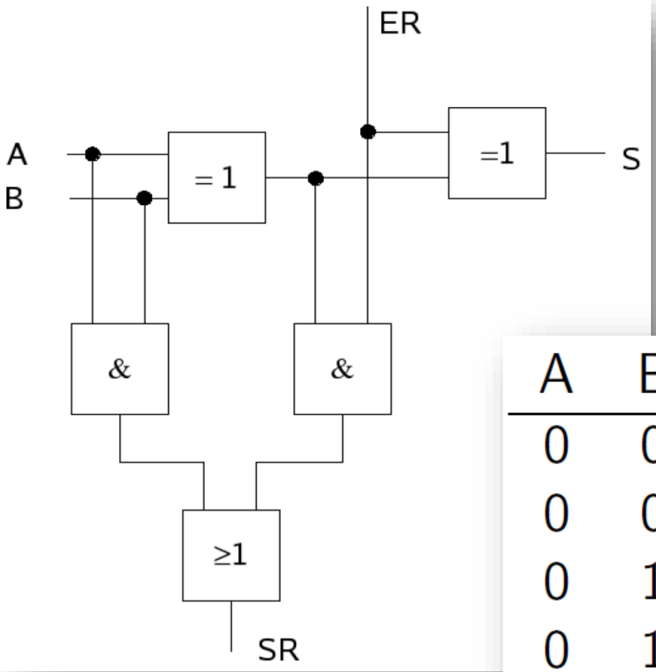


A	B	S	R
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1



A	B	ER	S	SR
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Combinatorial

Source: https://en.wikipedia.org/wiki/Barrel_shifter

3) LOGIC CIRCUITS

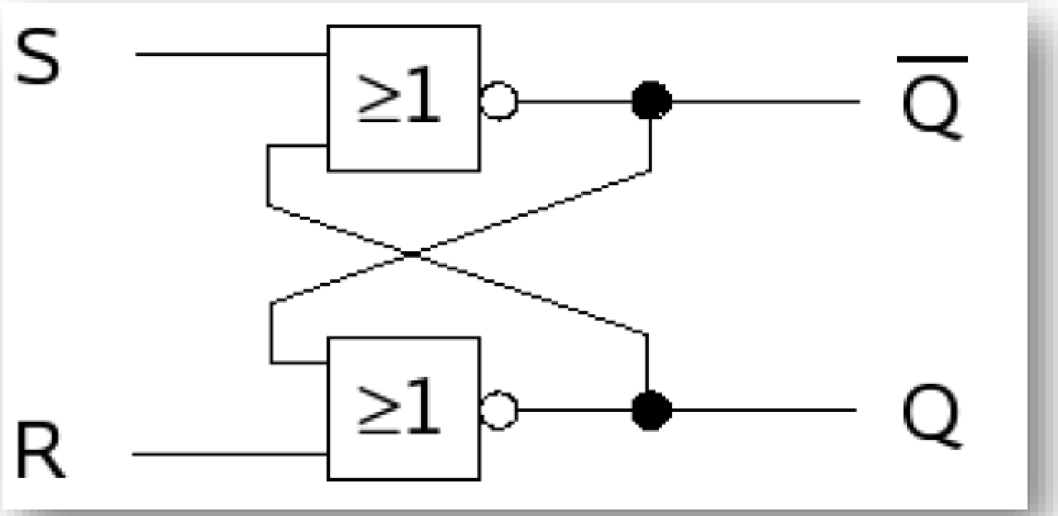
- Flip-flops and latches:
  - Sequential circuits to **memorize** one bit state

Sequential

Source: https://en.wikipedia.org/wiki/Flip-flop_(electronics)

3) LOGIC CIRCUITS

- SR NOR latch:
  - Set (S) to put output to 1
  - Reset (R) to put output to 0
  - Problem! When R and S are simultaneously at 1: unstable state => BAD!!!

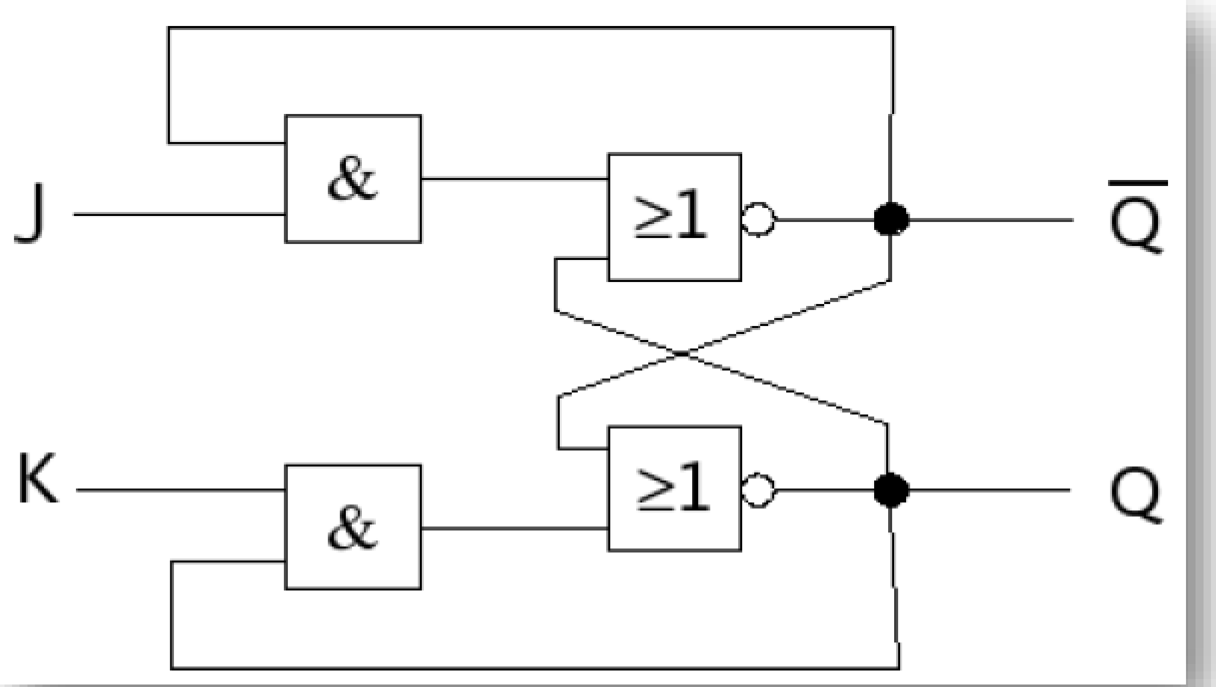


Sequential

Source: https://en.wikipedia.org/wiki/Flip-flop_(electronics)

3) LOGIC CIRCUITS

- JK latch:
  - No more instability but still two inputs

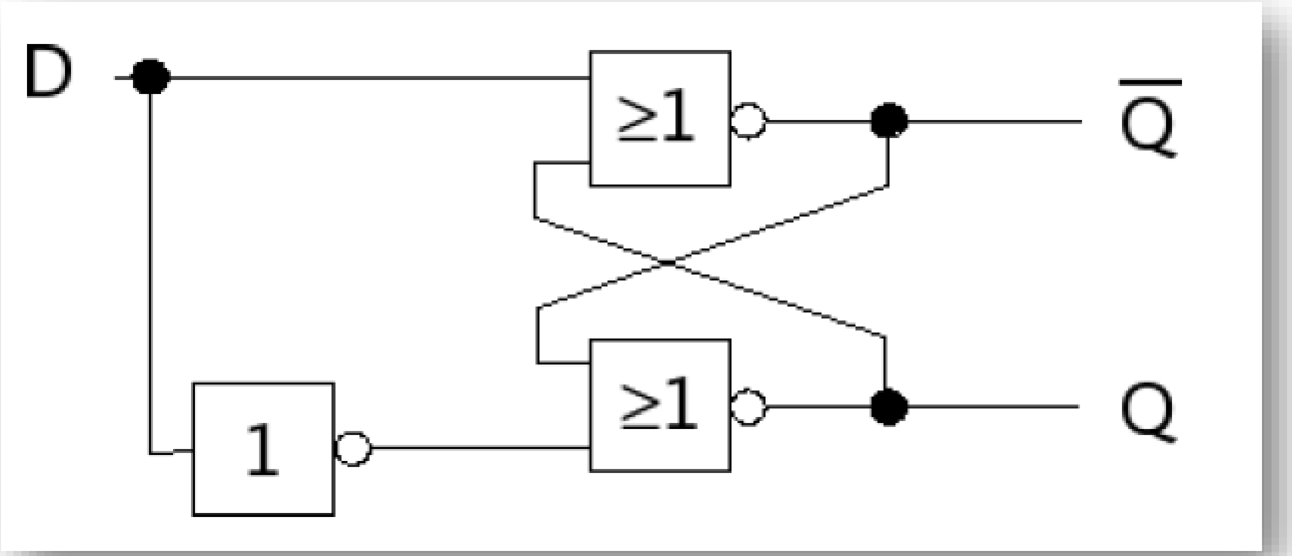


Sequential

Source: https://en.wikipedia.org/wiki/Flip-flop_(electronics)

3) LOGIC CIRCUITS

- D flip-flop:
  - No more instability and only one input

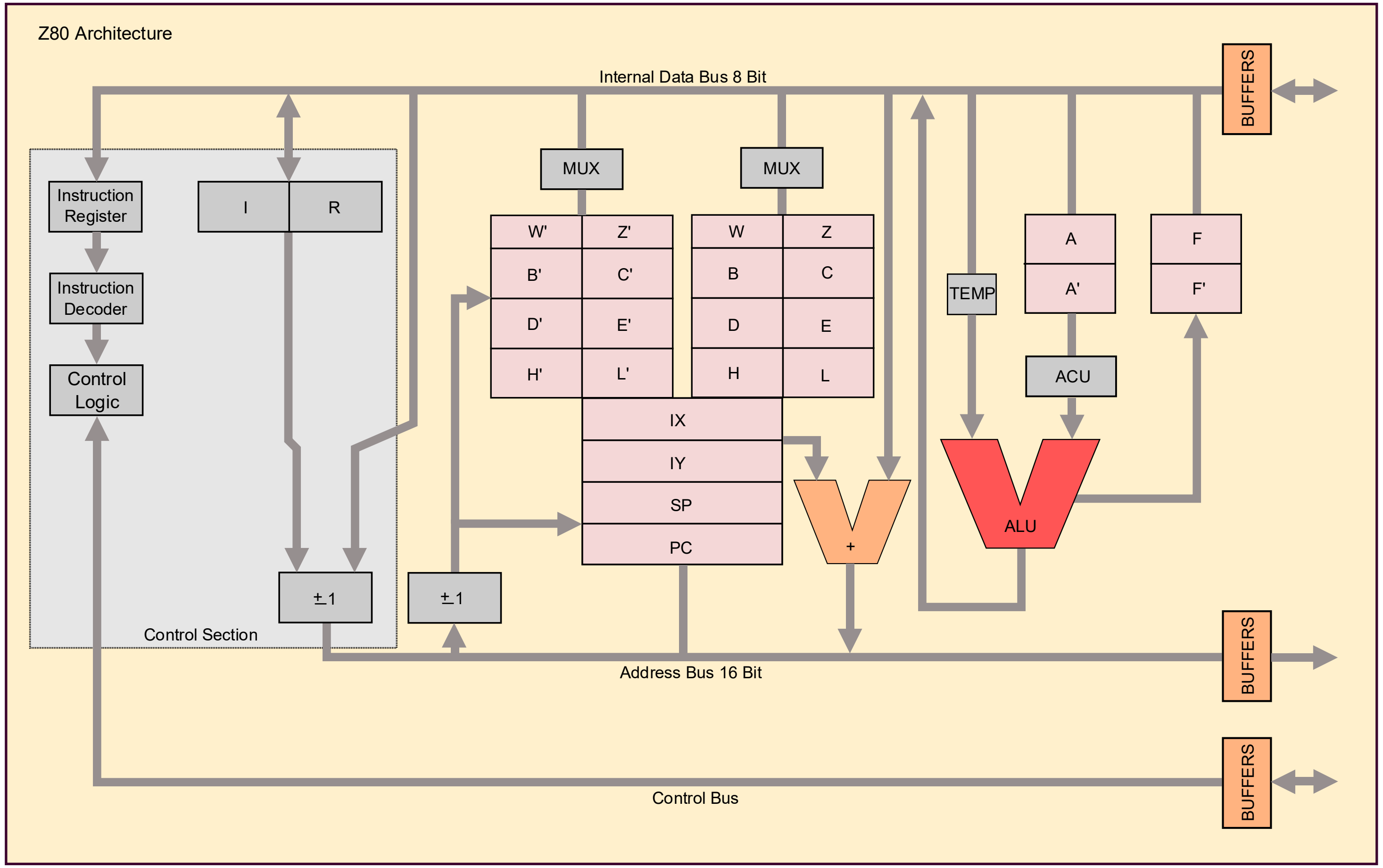


Sequential

Source: https://en.wikipedia.org/wiki/Flip-flop_(electronics)

4) MICROPROCESSORS

- A microprocessor is an electronic circuit which is able to memorize information internally (into registers), to compute operations (addition, subtraction, multiplication, division, comparison…) an to communicate using inputs/outputs with the external memory
- The clock ensures the synchronization of all the operations



Source: https://en.wikipedia.org/wiki/Microprocessor

ANY QUESTIONS?